CA-4: A/B Testing Report

IXD318 · User Data Analytics & Implementation

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1. Objective

This report documents an A/B usability test conducted on two visual variants of the ChatGPT mobile interface — Version A (Light Theme) and Version B (Dark Theme). The aim was to understand how design choices — specifically colour scheme and layout organisation — affect real users' comfort, navigation confidence, and overall preference. Five participants completed a set of structured tasks on both versions, and their responses were observed, recorded, and analysed to draw practical design conclusions.

2. Testing Method

How the test was run

Each participant was shown both versions in randomised order to minimise sequencing bias. They were asked to complete three lightweight tasks — starting a new chat, finding a previous conversation in the sidebar, and reading a poem response — while thinking aloud. A moderator observed silently and noted moments of hesitation, confusion, or confident action. After finishing both versions, participants completed a short verbal debrief covering preference, comfort, and any friction they noticed.

Tasks given to participants

#	Task	Success Criteria
Task 1	Start a new conversation	Tap the compose / new-chat button
Task 2	Find a previous chat	Navigate to a past session in the sidebar
Task 3	Read a response	Scroll through a poem reply and rate readability

3. Participants

Five participants were recruited — a mix of students and young professionals who use AI chat tools at least occasionally. Ages ranged from 19 to 26. All had some familiarity with ChatGPT, though none had used the specific interface variants shown.

ID	Age	Background	Device Familiarity	Prior ChatGPT Use
P1	21	Design student	High	Regular
P2	24	Engineering student	High	Occasional
P3	19	First-year undergrad	Medium	Occasional
P4	26	Working professional	Medium	Regular
P5	22	Arts student	Low	Rare

4. User Feedback

Below is a summary of what each participant said and did during the test. Direct observations are noted alongside paraphrased quotes from the debrief.

P1 · Preferred: Version B (Dark Theme)

What they said

"Dark mode is just easier on my eyes — I could read for longer without squinting. The light version felt a bit clinical."

Observed behaviour

Completed all 3 tasks confidently on both. Slight pause finding the sidebar in Version A.

P2 · Preferred: Version B (Dark Theme)

What they said

"I liked how the dark version groups the quick-action buttons — it felt more organised."

Observed behaviour

No errors on either. Noted that Version A felt busier on the home screen.

P3 · Preferred: Version A (Light Theme)

What they said

"I'm used to light mode apps, so the dark one confused me for a second. The white version felt more familiar."

Observed behaviour

Hesitated on Task 2 in Version B — took around 5 seconds longer to locate the sidebar toggle.

P4 · Preferred: Version B (Dark Theme)

What they said

"Dark mode for professional use just looks sleeker. It also felt less distracting."

Observed behaviour

Completed all tasks without issues. Appreciated the contrast of the dark input bar.

P5 · Preferred: No strong preference

What they said

"Both were pretty easy once I figured out the layout. Maybe dark is slightly nicer?"

Observed behaviour

Briefly confused by the new-chat button in Version A — mistook it for a settings icon.

5. Comparison Table

The table below compares both versions across the key dimensions observed during testing. Ratings are qualitative and based on participant behaviour and verbal feedback.

Dimension	Version A — Light	Version B — Dark
Visual comfort	Good in bright environments	Better for extended reading
Contrast / legibility	High contrast, clean	Strong contrast, text pops
Button discoverability	Moderate — slight confusion noted	Good — grouped actions clear
Sidebar navigation	Easy once found	Slightly slower for first-timers
Aesthetic appeal	Familiar, clinical	Modern, polished
Error/confusion rate	1 instance (new-chat icon)	1 instance (sidebar toggle)
Overall preference	1 out of 5 participants	3 out of 5 participants

6. Final Insights

Which version performed better — and why

Version B (Dark Theme) was the clear winner, preferred by 3 out of 5 participants, with a fourth expressing mild agreement. Here is what the test revealed:

> **Dark mode reduces eye strain**
>
> Three participants independently mentioned that the dark interface felt easier to read on a phone screen, especially for longer responses like the poem task. This aligns with broader UX research on OLED screens and ambient lighting conditions.

Grouped quick-action buttons in Version B aided discoverability

Version A placed shortcut buttons in a single horizontal row on the home screen, which felt slightly crowded. Version B's grid layout with icon labels was perceived as more organised and easier to scan.

Version A is still valid for certain user groups

P3 — the least tech-familiar participant — preferred the lighter interface because it matched existing mental models from other apps. This is an important reminder that dark mode is not universally preferred, particularly for novice or infrequent users.

Minor friction existed in both versions

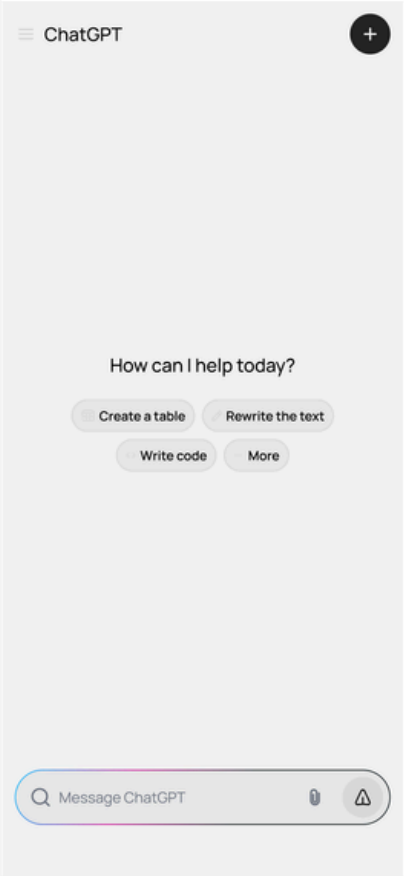
Version A caused one confusion with the new-chat icon (mistaken for settings). Version B caused a slight delay on sidebar navigation for first-time users. Neither error was serious, but both point to icon clarity as an area to refine.

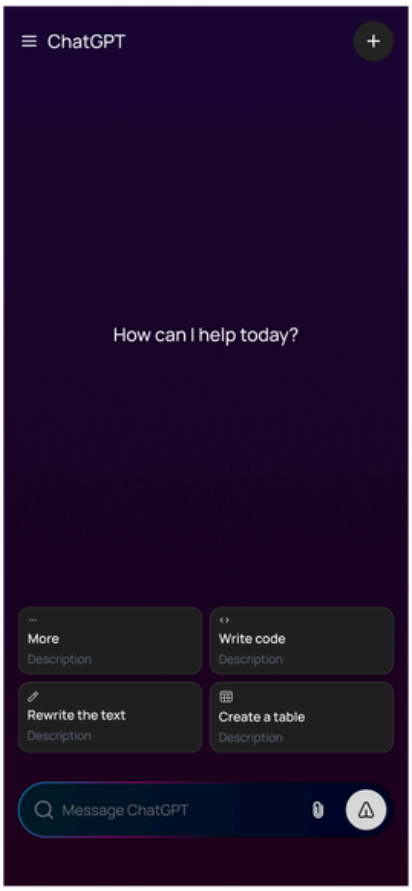
Recommendation

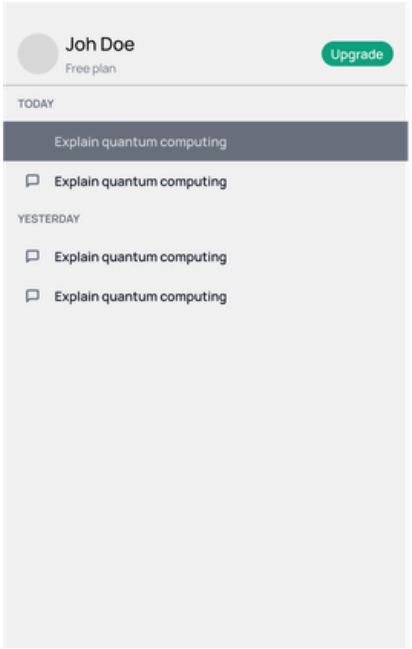
Maintain Version B as the default interface while keeping Version A as a light-mode toggle. Revisit icon labelling on both versions — particularly the compose button in Version A and the sidebar control in Version B. A future test with 10+ participants across age groups would strengthen these findings.

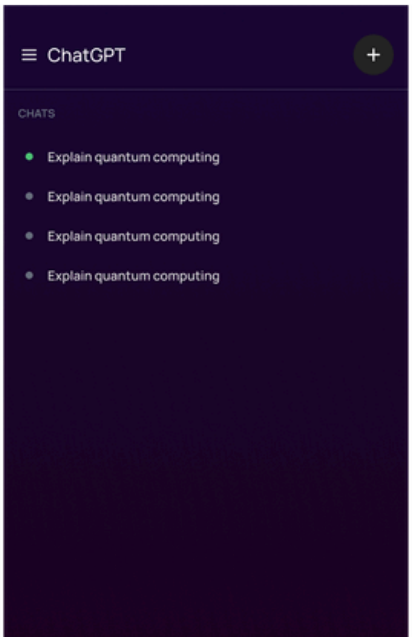
This test demonstrates the value of putting even polished-looking interfaces in front of real users. Both versions worked, but the dark theme created a noticeably more

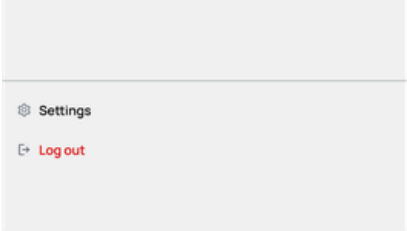
comfortable and confident experience for the majority. Small differences in layout and colour have a measurable impact on how people feel while using a product — and that feeling shapes whether they come back.

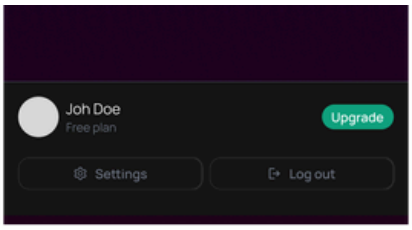

ChatGPT
How can I help today?
Create a table
Rewrite the text
Write code
More
Message ChatGPT

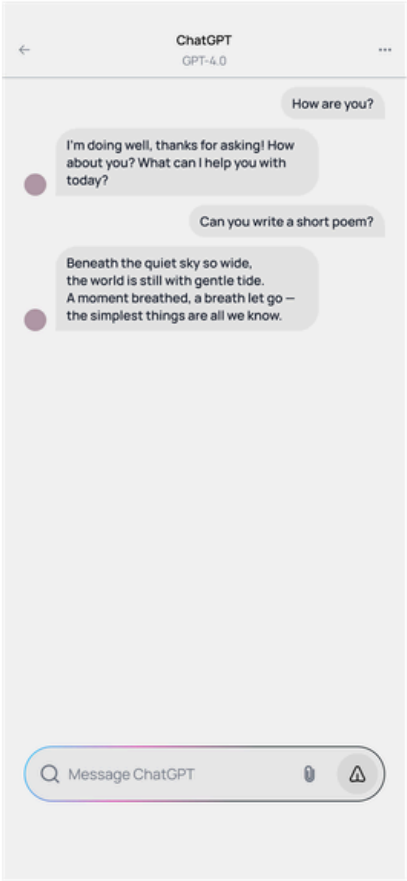

ChatGPT
How can I help today?
More
Description
Write code
Description
Rewrite the text
Description
Create a table
Description
Message ChatGPT

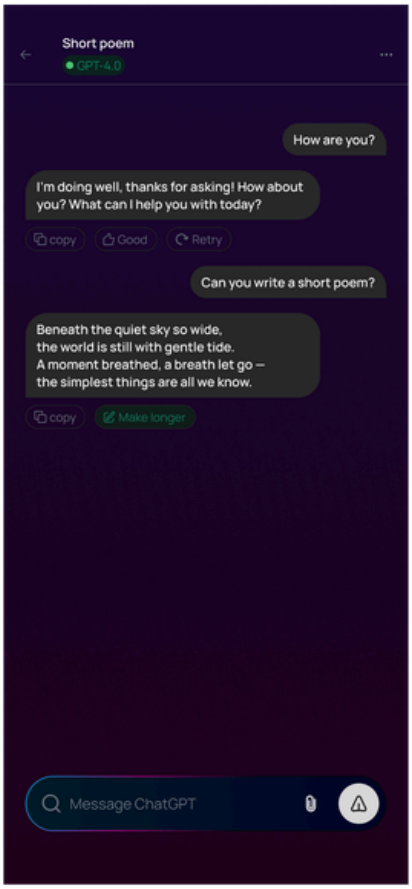

Joh Doe
Free plan
Upgrade
TODAY
Explain quantum computing
Explain quantum computing
YESTERDAY
Explain quantum computing
Explain quantum computing


ChatGPT
CHATS
Explain quantum computing
Explain quantum computing
Explain quantum computing
Explain quantum computing


Settings
Log out


Joh Doe
Free plan
Upgrade
Settings
Log out


ChatGPT
GPT-4.0
How are you?
I'm doing well, thanks for asking! How about you? What can I help you with today?
Can you write a short poem?
Beneath the quiet sky so wide,
the world is still with gentle tide.
A moment breathed, a breath let go —
the simplest things are all we know.
Message ChatGPT


Short poem
GPT-4.0
How are you?
I'm doing well, thanks for asking! How about you? What can I help you with today?
copy
Good
Retry
Can you write a short poem?
Beneath the quiet sky so wide,
the world is still with gentle tide.
A moment breathed, a breath let go —
the simplest things are all we know.
copy
Make longer
Message ChatGPT